Stationary Analysis of a Battery-Buffered EV Charging Station

A Piecewise-Deterministic Approach

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Outline

System Model

The PDMP Formulation

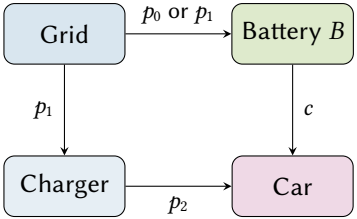
Exponential Demands

General Energy Demands

Summary and Open Problems

Why Battery-Buffered Charging?

- Fast EV charging requires high peak power from the grid
- Grid connection is limited — upgrading is costly
- A **battery buffer** absorbs excess demand
- Charges slowly from the grid, discharges fast to the car



Charging Rates

- $p_0 \leq p_1$ grid-to-battery charge rate when idle
  kept below p_1 to extend battery life
- p_1 grid rate when a car is present and battery is empty
- $p_2 > p_1$ fast charging rate when battery is positive
- $c = p_2 - p_1$ net battery discharge rate during fast charging
- B battery capacity

Ordering: $p_0 \leq p_1 < p_2$, so $c > 0$.

Customers and Service

- Cars arrive as a **Poisson process** of rate λ
- Each car has an energy demand $Y \sim F$, mean $1/\mu$
- **Loss system**: a car arriving when the charger is busy is turned away

Two-phase service

- **Phase 1** — battery positive ($x > 0$): car charges at rate p_2, battery depletes at rate c
- **Phase 2** — battery empty ($x = 0$): car charges at rate p_1, battery stays at 0

Piecewise-Deterministic Markov Processes

A PDMP [?] is defined by three local characteristics:

Flow $r(x)$: deterministic ODE $\dot{x} = r(x)$ between jumps

Jump rate $\lambda(x)$: spontaneous jumps occur at rate $\lambda(x)$

Kernel $Q(\cdot\,; x)$: post-jump location given pre-jump state x

Between jumps the process follows a deterministic trajectory.
At jump times it relocates randomly according to Q.

State Space

State: $Z(t) = (n(t),\ x(t)) \in \{0, 1\} \times [0, B]$

$n = 0$ charger idle — battery recharges from the grid

$n = 1$ car being served — battery discharges (if $x > 0$)

x battery level — continuous, bounded in $[0, B]$

Between jumps, $x(t)$ evolves deterministically according to $\dot{x} = r(n, x)$.
Jumps occur at Poisson-like times and flip n.

The Vector Field

n	condition	$\dot{x} = r(n, x)$	
0	$x < B$	$+p_0$	(battery charges)
0	$x = B$	0	(battery full, stays)
1	$x > 0$	$-c$	(battery discharges)
1	$x = 0$	0	(battery empty, stays)

Setting $r = 0$ at the boundaries avoids forced jumps.

Cost: we lose Lipschitz continuity at $x = 0$ and $x = B$.

Fix: uniqueness via a monotonicity argument — next slide.

Well-Posedness of the Flow

Question: does $\dot{x} = r(n, x)$ have a unique solution without Lipschitz?

Uniqueness via Monotonicity

If $r(n, \cdot)$ is *non-increasing*, then the ODE has at most one solution.

Proof sketch: Let $d(t) = x(t) - y(t)$ for two solutions. Then

$$\frac{d}{dt}\, d(t)^2 = 2\, d(t)\, [r(n, x(t)) - r(n, y(t))] \leq 0$$

since r is non-increasing. With $d(0) = 0$ this gives $d(t) = 0$ for all $t \geq 0$.

In our model $r(0, \cdot)$ drops from p_0 to 0 at $x = B$, and $r(1, \cdot)$ drops from $-c$ to 0 at $x = 0$ — both non-increasing. Existence follows from Carathéodory.

Jumps and Generator

Spontaneous jumps only (no forced boundary jumps):

- Arrival at $(0, x)$: rate λ, flip n: $0 \to 1$, x unchanged
- Departure at $(1, x)$, $x > 0$: rate μp_2, flip n: $1 \to 0$, x unchanged
- Departure at $(1, 0)$: rate μp_1, flip n: $1 \to 0$

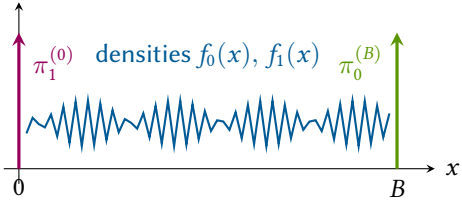
Generator $\mathcal{A}$ acting on test functions ψ:

$$\mathcal{A}\psi(0, x) = r(0, x)\, \psi_0'(x) + \lambda\, [\psi(1, x) - \psi(0, x)]$$
$$\mathcal{A}\psi(1, x) = r(1, x)\, \psi_1'(x) + \mu p(x)\, [\psi(0, x) - \psi(1, x)]$$

where $p(x) = p_2\, \mathbf{1}_{x>0} + p_1\, \mathbf{1}_{x=0}$.

Structure of the Stationary Distribution

The stationary distribution π has a **mixed** structure:



- $\pi_1^{(0)}$: probability of battery empty, car being served
- $\pi_0^{(B)}$: probability of battery full, charger idle
- No mass at $(0, 0)$: battery always positive at arrivals (Poisson)

Kolmogorov Forward Equations

Setting $\partial/\partial t = 0$ in the forward equation gives the **interior balance**:

$$p_0 f_0'(x) = -\lambda f_0(x) + \mu p_2 f_1(x) \tag{E0}$$

$$-c f_1'(x) = \lambda f_0(x) - \mu p_2 f_1(x) \tag{E1}$$

Boundary flux balance at $x = 0$ and $x = B$:

$$p_0 f_0(0^+) = c f_1(0^+) = \mu p_1 \pi_1^{(0)}$$

$$p_0 f_0(B^-) = c f_1(B^-) = \lambda \pi_0^{(B)}$$

Each equation says: *probability flux in = probability flux out.*

Proportionality and the Key ODE

Step 1. Add (E0) and (E1):

$$\frac{d}{dx}\left[p_0 f_0(x) - c f_1(x)\right] = 0$$

So $p_0 f_0 - c f_1 = \text{const}$. The boundary condition at $x = 0$ gives const $= 0$:

$$\boxed{f_1(x) = \frac{p_0}{c} f_0(x)}$$

Step 2. Substitute into (E0):

$$f_0'(x) = \underbrace{\left(\frac{\mu p_2}{c} - \frac{\lambda}{p_0}\right)}_{\theta_1} f_0(x)$$

The solution is $f_0(x) = f_0(0^+)\, e^{\theta_1 x}$.

Physical Meaning of θ_1

Consider the battery level *without* the constraints at 0 and B:

- Battery charges at rate p_0 when idle (fraction of time $\approx \mu p_2/(\lambda + \mu p_2)$)
- Battery drains at rate c when busy (fraction of time $\approx \lambda/(\lambda + \mu p_2)$)

The **mean drift** of the unconstrained battery is proportional to:

$$\theta_1 = \frac{\mu p_2}{c} - \frac{\lambda}{p_0}$$

$\theta_1 > 0$	battery drifts up — station lightly loaded
$\theta_1 = 0$	critical balance — uniform density
$\theta_1 < 0$	battery drifts down — station heavily loaded

Complete Solution

$$f_0(x) = \frac{e^{\theta_1 x}}{K}, \qquad f_1(x) = \frac{p_0}{c} \cdot \frac{e^{\theta_1 x}}{K}, \qquad x \in (0, B)$$

$$\pi_1^{(0)} = \frac{p_0}{\mu p_1 K}, \qquad \pi_0^{(B)} = \frac{p_0}{\lambda K} e^{\theta_1 B}$$

Normalization constant:

$$K = \frac{p_0 + c}{c} \cdot \frac{e^{\theta_1 B} - 1}{\theta_1} + \frac{p_0}{\mu p_1} + \frac{p_0}{\lambda} e^{\theta_1 B}$$

A Key Structural Result

Decoupling in the interior

The battery level and the charger occupancy are independent given $x \in (0, B)$:

$$\Pr(n = 1 \mid x) = \frac{p_0}{p_0 + c} \qquad \text{for all } x \in (0, B)$$

Knowing the battery level gives no information about whether a car is being served.

Performance Metrics

Metric	Formula	Limit $B \to \infty$
Blocking P_B	$\frac{p_0}{K}\left(\frac{e^{\theta_1 B}-1}{c\,\theta_1}+\frac{1}{\mu p_1}\right)$	$\frac{\lambda}{\lambda+\mu p_2}$
Mean efficiency	$\frac{1}{p_2}+\left(\frac{1}{p_1}-\frac{1}{p_2}\right)\cdot(\cdots)$	$\frac{1}{p_2}$
Fast charging ρ_{fast}	$\frac{p_0}{cK}\cdot\frac{e^{\theta_1 B}-1}{\theta_1}$	$P_B-\rho_{\text{slow}}$
Slow charging ρ_{slow}	$\frac{p_0}{\mu p_1 K}$	0

Performance Metrics — Limits

As $B \to 0$

Blocking $\to \dfrac{\lambda}{\lambda + \mu p_1}$ and efficiency $\to \dfrac{1}{p_1}$ (M/M/1/1 at slow rate)

As $B \to \infty$

Blocking $\to \dfrac{\lambda}{\lambda + \mu p_2}$ and efficiency $\to \dfrac{1}{p_2}$ (M/M/1/1 at fast rate)

All metrics are explicit functions of $(p_0, p_1, p_2, \mu, \lambda, B)$.

What Breaks for General F

For $Y \sim \text{Exp}(\mu)$: the **residual demand** at any time is again $\text{Exp}(\mu)$.

$\Rightarrow$ departure rate depends only on (n, x) $\Rightarrow$ (n, x) is Markov.

For general F: the departure rate depends on how long the car has been served and at what rate — which depends on the battery trajectory.

$\Rightarrow$ (n, x) is **not** Markov.

Fix: augment the state

Track the **attained service** $s(t)$ = energy already delivered.
Departure rate: $h(s)$ = hazard rate of F evaluated at attained service.
This gives only **spontaneous jumps** — no forced boundary behaviour.

Augmented PDMP

State: $Z(t) = (n(t),\ x(t),\ s(t))$ with $s = 0$ when $n = 0$

Flow:

$$(\dot{x}, \dot{s}) = \begin{cases} (p_0,\ 0) & n = 0,\ x < B \\ (-c,\ p_2) & n = 1,\ x > 0 \\ (0,\ p_1) & n = 1,\ x = 0 \end{cases}$$

Jumps:

- Arrivals at rate λ: $(0, x, 0) \to (1, x, 0)$, fresh demand begins
- Departures at rate $h(s)$: $(1, x, s) \to (0, x, 0)$, s resets

The characteristics in the (x, s) plane are lines $p_2 x + cs = \text{const.}$

Solution by Characteristics

The forward equation for $f_1(x, s)$ is a transport PDE:

$$-c\,\partial_x f_1 + p_2\,\partial_s f_1 = -h(s)\,f_1$$

Integrating along characteristics from the entry boundary $s = 0$:

Characteristic solution

$$f_1(x, s) = \frac{\lambda}{p_2} f_0\left(x + \frac{cs}{p_2}\right)\bar{F}(s), \qquad s \leq \frac{p_2(B - x)}{c}$$

and $f_1(x, s) = 0$ for $s > p_2(B - x)/c$.

Interpretation: the density at (x, s) comes from cars that arrived at battery level $x + cs/p_2$ with energy $Y > s$ still unserved.

Closing the Equation — Connection to Abandonment Queues

Substituting the characteristic solution into the equation for f_0:

Volterra integro-differential equation

$$p_0 f_0'(x) = -\lambda f_0(x) + \frac{\lambda}{c} \int_x^B f_Y\left(\frac{p_2(u-x)}{c}\right) f_0(u)\, du$$

This is the same equation that governs the M/G/1+G queue — an M/G/1 system where customers abandon with a general patience distribution. The kernel $f_Y(p_2(u-x)/c)/c$ plays the role of the patience density.

- No closed form for general F and finite B
- Tractable cases: exponential F, phase-type F, $B = \infty$ with $\theta_1 < 0$

What Remains Tractable

Case	Method	Result
Exp(μ), finite B	ODE	Closed form
Phase-type, finite B	ODE system	Sum of exponentials
General F, $B = \infty$, $\theta_1 < 0$	Laplace transform	$\hat{f}_0(s) = p_0 f_0(0^+)/\psi(s)$
General F, finite B	Volterra equation	Numerical

Distribution-free results (hold for any F with mean $1/\mu$):

- Stability condition $\theta_1 < 0$ — depends only on $E[Y]$
- Blocking P_B — depends only on $E[Y]$
- Mean battery $E[x]$ — depends on $E[Y]$ and $E[Y^2]$
- Charging efficiency $E[T]/E[Y]$ — depends on full stop-loss transform of F

Summary and Open Problems

Summary

- Battery-buffered EV station modelled as a PDMP
- Well-posedness via monotonicity (no Lipschitz needed)
- Exponential demands: exact closed-form stationary distribution
- Key structural result: charger occupancy $\perp$ battery level in interior
- General demands: Volterra equation, connection to M/G/1+G

Open Problems

- Multi-server extension ($s > 1$, shared grid)
- Waiting room (non-loss) systems
- Optimal sizing of B, p_0, p_2
- Heavy-traffic limits
- Numerical methods via Volterra solvers
- Connection to M/G/1+G: can abandonment results transfer?